Amazon Virtual Pri

(Amazon VPC)

Public v4 Ip version have 4 billion ip

to move above IPv6 comes with 2^{32}

IPv6 is 128 bit so its hard to remember and not supported by everyone

In VPC it works with Private IPs

Private Ip → gateway → Public Ip

CIDR

192.168.1.0/24 → subnet mask

fixed	flexible

IP range = 192.168.0 – 255

and subnet mask of 255.255.255.0

Private

→ 10.x.x.x

→ 172.16.x.x — 172.31.x.x

→ 192.168.x.x

NACL (Network access control list)

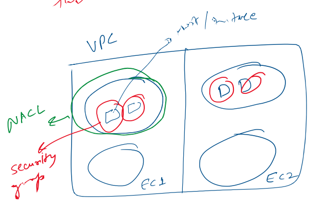
→ Inbound and Outbound

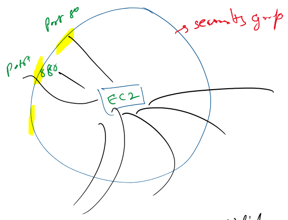
→ Lowest number have high priority

→ Default permission is Allow In and Allow out

→ when you create custom NACL ... default DENY RULE

as ... it will ...
then it ...





- same security group can be applied to multiple instances
→ Public subnet and Private subnet ~~the~~ customer has to create

Storage Module 4

3 types
- Block storage — ex laptop
- file storage — ex. NFS, SMB, CIFS
- object storage — ex. Atomic file which cannot be partial

In S3 5TB is size for objects

object can be partially updated

- Instance store are on NVME not on RAM
- Instance store doesnot have snapshot
-

Amazon EFS

→ it is outside Availability zone
→ It is regional service
→ It is for Linux or MAC
→ for windows, Amazon FSx is there

→ There are 4 types of FSx
you can search "help me for"
you will get more details.

Object Storage

S3
objects keys
Buckets Regions

→ S3 block public Access default
→ default S3 encryption [client side, server side]

storage classes in S3

- S3 standard
- " " infrequent Access
- S3 one zone " "
- Glacier [more cheaper]

... storage

Glacier

change attribute of file
and it cost 2$ per month

Amazon S3 versioning

if object deleted by mistake
then it create new version id

-> you can save storage by using
lifecycle so it will save 3
last versions

-> Disabled by default

-> when you enable version
then it will not delete it will
stop creating versions and
put it in suspend

-> if you delete the delete marker
in versioning then it will
again there

-> if you delete object then it will add
add delete marker on object
and it show like it deleted

Databases

Benifits of databases on AWS
- -> purpose built
- -> performance at scale
- -> Fully managed
- -> Secure and highly available

Data types
- -> Relational
- -> key value
- -> In memory [Redis , mem cached
- -> Document [mongo db
- -> wide colum [cassandra)
- -> Graph [neptune] used to store relation eg social network fb
- -> Time Series [Time stamp]
- -> Ledger [block chain]

Relational db

-> having Relations

Benifits
- -> Complex SQL
- -> Reduced redundancy
- -> Familiarity
- -> ACID principle

ON Prem Vs EC2 instance

[Image captured] 7:03pm

Managed db in AWS] RDS

Img captured 7:05

Amazon RDS

Img 7:08
7:08

Standard -> db.m6g
memory optimized -> db.x2g db.r

Burstable performance -> db.t4
db.t3
db.t2

DB Instance Storage
- -> General purpose 16k
- -> Provision IOPS 64k
- -> Magnetic

-> RDS is not encrypted
by default

-> create RDS on private
Subnet

DB backup

img 7:15

DB Redundancy

Amazon Dynodb

-> its key value pair
-> Encryption on rest and default
encrypted

-> Any scale
-> less than 10 milli second
->

Amazon

Auto

Pa / sql

- copy 3 places
- asynchronous.

MONITORING

Amazon Cloud Watch

Alarms

Load balancing

Elastic load balancing [ELB]

ELB Types

- Application load balancer
  - ex.com/mobile
  - ex.com/regist
- Network LB (Layer 4) TCP UDP
  - ~~LAB~~ NLB
- Gateway LB [virtual appliances
- ~~Classic~~ classic LB
  - slow, cheaper

Scaling

- vertical → scale up and down (32 x 2 map size)
- Horizontal → scale out / in

Horizontal is preferred

EC2 Auto Scaling

- Improve fault tolerance
- Increase application availability
- Lower cost

Draining:
- Stop taking new connections
- This will save when old instances are going to delete

Launch template

Amazon EC2 Auto Scaling group

Scaling policies

- Simple scaling policy
- Step "
- Target tracking scaling policy

Auto Scaling

- Automatic
- Scheduled
- Fleet management
- Predictive [pattern based
- multiple purchase options
- Included with Amazon EC2